

GILLESPIE COUNTY AP, TX, USA

WMO: 720323

Lat: 30.243N

Lon: 98.910W

Elev: 1695

StdP: 13.82

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 93947

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.4	27.4	7.7	8.7	39.7	12.3	11.0	41.7	24.4	49.8	21.0	52.5	4.6	0	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.8	97.4	69.5	95.4	69.7	93.3	70.2	75.8	83.6	74.9	83.1	74.0	82.8	8.3	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.3	132.0	79.4	72.6	128.7	78.7	72.0	126.0	78.1	40.7	85.9	39.5	81.9	38.7	82.7	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.1	18.0	16.2	DB	17.7	101.3	3.8	2.4	15.0	103.0	12.7	104.4	10.6	105.7	7.8	107.4	
			WB	15.9	76.4	3.4	1.9	13.5	77.8	11.5	78.9	9.6	79.9	7.1	81.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.1	47.9	52.3	59.4	65.9	72.6	79.2	81.6	82.3	76.6	67.3	57.2	49.5
	DBStd	14.09	8.95	10.10	8.62	7.22	6.21	3.63	3.04	3.36	4.98	8.09	9.19	8.83
	HDD50	416	146	86	22	2	0	0	0	0	0	3	38	119
	HDD65	2019	534	365	210	76	15	0	0	0	1	72	262	484
	CDD50	6277	80	150	314	479	700	875	981	1001	799	539	254	105
	CDD65	2403	3	8	37	103	250	425	516	536	350	143	28	4
	CDH74	24221	33	99	286	868	2077	4280	5668	6168	3310	1205	183	44
	CDH80	10715	1	12	42	220	707	1918	2842	3218	1414	319	20	2
(14) Wind	WSAvg	6.5	6.4	7.0	8.0	8.1	7.5	6.9	6.0	5.4	4.7	5.8	6.2	6.1
(15) Precipitation	PrecAvg	33.6	1.9	2.1	3.0	2.6	3.7	3.6	3.1	2.5	3.3	3.4	2.5	2.0
	PrecMax	54.9	4.4	6.8	9.5	7.8	9.1	12.0	21.8	8.9	7.2	7.9	7.6	13.5
	PrecMin	13.8	0.2	0.0	0.8	0.2	0.7	0.2	0.2	0.1	0.3	0.1	0.0	0.0
	PrecStd	10.1	1.4	1.8	1.9	1.9	2.1	2.9	4.7	2.2	1.9	2.2	2.2	2.8
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.5	81.2	84.3	90.7	93.4	98.5	99.2	99.5	97.1	90.2	83.0	79.1
		MCWB	55.1	57.7	60.0	63.1	69.7	69.9	68.9	69.7	67.8	67.0	64.6	58.6
	2%	DB	72.9	77.1	80.4	85.6	90.2	94.8	96.9	97.9	93.0	86.4	79.1	73.4
		MCWB	57.2	57.7	61.5	64.1	69.4	70.0	69.1	70.1	69.6	67.7	63.6	60.0
	5%	DB	68.7	72.5	77.2	81.8	87.1	91.8	94.8	95.7	90.7	83.8	75.4	70.4
		MCWB	55.7	58.3	60.9	64.8	69.1	70.2	69.7	70.1	69.0	67.7	62.7	57.7
	10%	DB	64.3	68.9	73.4	79.2	83.9	89.9	92.7	93.5	88.1	81.0	72.3	65.8
		MCWB	54.2	58.5	61.0	63.7	68.4	70.2	70.2	70.5	68.8	65.8	62.1	57.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.3	66.3	70.1	72.0	75.5	77.0	76.4	76.9	76.3	74.4	70.2	68.0
		MCDB	70.0	71.7	73.5	78.3	83.0	85.1	83.5	87.1	83.7	80.4	74.8	74.0
	2%	WB	62.8	64.5	67.3	70.5	74.0	75.3	75.2	75.4	74.8	72.8	68.6	64.4
		MCDB	67.6	69.7	74.1	77.2	82.7	83.6	82.8	83.9	82.2	78.9	74.1	68.7
	5%	WB	59.8	62.8	65.5	69.1	72.5	74.2	74.2	74.4	73.4	71.3	66.6	62.1
		MCDB	64.6	67.5	72.1	76.2	80.7	83.3	82.6	83.7	81.7	78.1	73.0	66.3
	10%	WB	56.1	60.5	64.0	67.8	71.3	73.0	73.3	73.4	72.3	69.9	64.3	58.7
		MCDB	62.3	66.0	69.9	75.2	78.8	82.6	82.3	83.5	80.5	76.7	70.0	63.5
(35) Mean Daily Temperature Range	5% DB	MDBR	24.0	23.4	23.1	22.9	19.9	19.9	20.8	21.8	21.1	22.8	22.6	23.7
		MCDBR	30.9	28.1	25.8	25.8	23.4	23.8	24.6	25.0	24.4	23.0	24.6	30.1
		MCWBR	17.5	15.6	12.7	11.7	8.6	6.4	5.7	5.6	7.1	9.6	13.2	17.3
	5% WB	MCDBR	22.1	20.3	19.4	19.2	17.5	18.0	18.2	19.8	17.6	17.2	19.6	21.1
		MCWBR	16.5	14.7	12.1	10.7	8.3	6.4	5.5	6.0	6.3	9.2	13.0	15.9
(40) Clear-Sky Solar Irradiance	taub		0.308	0.315	0.337	0.379	0.409	0.409	0.418	0.408	0.389	0.357	0.334	0.308
	taud		2.534	2.518	2.467	2.368	2.333	2.385	2.399	2.409	2.408	2.486	2.486	2.537
	Ebn at Noon		295	303	301	290	280	279	275	278	280	284	283	289
	Edh at Noon		27	30	34	39	41	38	38	37	36	31	29	26
(44) All-Sky Solar Radiation	RadAvg		997	1188	1491	1777	1867	2145	2135	2020	1657	1363	1045	889
	RadStd		109	183	140	166	200	157	181	113	159	195	132	118

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+140

Nomenclature: See separate page